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천체역학 2021 중간고사.

21.10.18 (화) 13시 응시.

$r = 7000 \text{ km}, \lambda = 0, F = 0$

$\dot{r} = 0, \dot{\lambda} = 0$   
 $\ddot{r} = 0, \ddot{\lambda} = 0$

$$\begin{cases} \ddot{r} - r\dot{\lambda}^2 \cos^2 \lambda - r\dot{\lambda}^2 = -\frac{M}{r^2} + \frac{Fr}{m} \\ r\ddot{\lambda} \cos \lambda + 2\dot{r}\dot{\lambda} \cos \lambda - 2r\dot{\lambda} \sin \lambda = \frac{F\lambda}{m} \\ r\ddot{\lambda} + 2\dot{r}\dot{\lambda} + r\dot{\lambda}^2 \sin \lambda \cos \lambda = \frac{F\lambda}{m} \end{cases}$$

$$\begin{cases} 0 - r\dot{\lambda}^2 - 0 = -\frac{M}{r^2} + 0 \rightarrow \dot{\lambda}^2 = \sqrt{\frac{M}{r^2}} \\ r\ddot{\lambda} + 0 - 0 = 0 \rightarrow \ddot{\lambda} = 0 \\ 0 + 0 + 0 = 0 \rightarrow 0 = 0 \end{cases}$$

$$\begin{aligned} \mathcal{D} &= \omega t + C \\ \dot{\mathcal{D}} &= \dot{\lambda} = 1.08 \times 10^{-3} \\ \ddot{\mathcal{D}} &= 0 \end{aligned}$$

$$\begin{aligned} V &= r\dot{\lambda} = 7000 \times 1.08 \times 10^{-3} = 7.56 \text{ km/s} \\ T &= P = \frac{2\pi}{\dot{\lambda}} = \frac{2\pi}{1.08 \times 10^{-3}} = 5814 \text{ s} = 1.6 \text{ hr} \end{aligned}$$

$r = R + \Delta r, \mathcal{D} = \mathcal{D}^* + \Delta \mathcal{D}, \lambda = 0 + \Delta \lambda$   
준 놓고 선형화 시작.

$\mathcal{D}^* = \omega t + C$   
 $\dot{\mathcal{D}}^* = \dot{\lambda}$   
 $\ddot{\mathcal{D}}^* = 0$

(i)  $(\ddot{r} + \Delta \ddot{r}) - (R + \Delta r)(\dot{\lambda}^* + \Delta \dot{\lambda})^2 \cos^2 \lambda - (R + \Delta r)\Delta \dot{\lambda}^2 = -\frac{M}{(R + \Delta r)^2} + \frac{Fr}{m}$

$\Delta \ddot{r} - (R + \Delta r)(\dot{\lambda}^{*2} + 2\dot{\lambda}^* \Delta \dot{\lambda} + \Delta \dot{\lambda}^2) \cdot 1 = -\frac{M}{R^2(1 + \frac{\Delta r}{R})^2} + \frac{Fr}{m}$

$\Delta \ddot{r} - (R\dot{\lambda}^2 + 2R\dot{\lambda}\Delta \dot{\lambda} + \Delta r\dot{\lambda}^2 + 2\Delta r\dot{\lambda}\Delta \dot{\lambda}) = -\frac{M}{R^2(1 + \frac{2\Delta r}{R})} = -\frac{M}{R^2}(1 + \frac{2\Delta r}{R})^{-1} = -\frac{M}{R^2}(\frac{R}{R} - \frac{2\Delta r}{R}) = -\frac{M}{R^3}(R - 2\Delta r) = -\dot{\lambda}^2 R + 2\dot{\lambda}^2 \Delta r + \frac{Fr}{m}$

$\Delta \ddot{r} - R\dot{\lambda}^2 - 2R\dot{\lambda}\Delta \dot{\lambda} - \dot{\lambda}^2 \Delta r = -\dot{\lambda}^2 R + 2\dot{\lambda}^2 \Delta r + \frac{Fr}{m} \Rightarrow \Delta \ddot{r} - 2R\dot{\lambda}\Delta \dot{\lambda} - 3\dot{\lambda}^2 \Delta r = \frac{Fr}{m}$

(ii)  $r\ddot{\lambda} \cos \lambda + 2\dot{r}\dot{\lambda} \cos \lambda - 2r\dot{\lambda} \sin \lambda = \frac{F\lambda}{m}$

$(R + \Delta r)\Delta \ddot{\lambda} \cos \lambda + 2(\dot{R} + \Delta \dot{r})(\dot{\lambda}^* + \Delta \dot{\lambda}) \cos \lambda - 2(R + \Delta r)(\dot{\lambda}^* + \Delta \dot{\lambda})(\Delta \dot{\lambda}) \sin \lambda = \frac{F\lambda}{m}$   
 $= R\Delta \ddot{\lambda} + 2\Delta \dot{r}\dot{\lambda} - 0 = \frac{F\lambda}{m} \Rightarrow R\Delta \ddot{\lambda} + 2\dot{\lambda}\Delta \dot{r} = \frac{F\lambda}{m}$

(iii)  $r\ddot{\lambda} + 2\dot{r}\dot{\lambda} + r\dot{\lambda}^2 \sin \lambda \cos \lambda = \frac{F\lambda}{m}$

$(R + \Delta r)\Delta \ddot{\lambda} + 2(\dot{R} + \Delta \dot{r})\dot{\lambda} + (R + \Delta r)(\dot{\lambda}^* + \Delta \dot{\lambda})^2 \Delta \lambda$   
 $= R\Delta \ddot{\lambda} + 2\dot{R}\Delta \dot{\lambda} + 2\dot{\lambda}\Delta \dot{r} + (R + \Delta r)\Delta \lambda(\dot{\lambda}^2 + 2\dot{\lambda}\Delta \dot{\lambda})$   
 $= R\Delta \ddot{\lambda} + R\Delta \lambda(\dot{\lambda}^2 + 2\dot{\lambda}\Delta \dot{\lambda}) + \Delta \dot{\lambda}\Delta \lambda(\dot{\lambda}^2 + 2\dot{\lambda}\Delta \dot{\lambda})$   
 $\Rightarrow R\Delta \ddot{\lambda} + R\dot{\lambda}^2 \Delta \lambda = \frac{F\lambda}{m}$

$$\begin{aligned} \Delta \ddot{r} - 2R\dot{\lambda}\Delta \dot{\lambda} - 3\dot{\lambda}^2 \Delta r &= \frac{Fr}{m} \\ R\Delta \ddot{\lambda} + 2\dot{\lambda}\Delta \dot{r} &= \frac{F\lambda}{m} \\ R\Delta \ddot{\lambda} + R\dot{\lambda}^2 \Delta \lambda &= \frac{F\lambda}{m} \\ x = \Delta r, y = R\Delta \dot{\lambda}, z = R\Delta \lambda \end{aligned}$$

$$\begin{aligned} \ddot{x} - 2\dot{y} - 3\dot{\lambda}^2 x &= \frac{Fr}{m} \\ \ddot{y} + 2\dot{x} &= \frac{F\lambda}{m} \\ \ddot{z} + \dot{\lambda}^2 z &= \frac{F\lambda}{m} \end{aligned}$$

(CW) 7.

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지구정지궤도  $r=42000$ ,  $\vec{w} = w \vec{k}$

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$$(\gamma \gamma) = (-110, 0, 10) \text{ km/s} = (x_0, y_0, z_0)$$

$$\dot{z} = z_h = 0.200 \text{ km/s} \quad w = \sqrt{\frac{400,000}{42000}} = 0.35 \text{ km/s} \quad (\dot{x} \dot{y} \dot{z}) = (-0.01, 0.02, 0.00) \text{ km/s} = (\dot{x}_0, \dot{y}_0, \dot{z}_0)$$



$$wt = 0.53$$

$$\sin wt = 0.5055$$

$$\cos wt = 0.8628$$

$$\tan wt = 0.5860$$

$$\begin{aligned} x(x_0 \dot{y}_0 - \dot{x}_0 y_0) + y(\dot{z}_0 x_0 - x_0 \dot{z}_0) + z(\dot{x}_0 y_0 - y_0 \dot{x}_0) &= 0 \\ y(\dot{z}_0 x_0 - x_0 \dot{z}_0) &= 0 \\ z(\dot{x}_0 y_0 - y_0 \dot{x}_0) &= 0 \end{aligned}$$

$$\dot{y}_0 = \frac{[6x_0(wt - \sin wt) - \dot{y}_0 \cdot w \cdot \sin wt - 2wx_0(4 - 3\cos wt)(1 - \cos wt)]}{(4\sin wt - 3wt)\sin wt + 4(1 - \cos wt)^2} = \frac{[-660(0.53 - 0.5055) - 0.01 \cdot 0.35 \cdot 0.5055 - 2 \cdot 110 \cdot 0.35 \cdot 0.5055 \cdot 0.14]}{0.45 \cdot 0.51 + 4 \cdot 0.14^2}$$

$$= \frac{-3.1187 \times 10^{-3} + 3.21457 \times 10^{-3}}{0.368} = 3.1135065 \times 10^{-3} = 0.00031$$

$$-0.00024 = \dot{y}_0^+$$

$$= \frac{[-660(0.53 - 0.5055) - 0.01 \cdot 0.35 \cdot 0.5055 - 2 \cdot 110 \cdot 0.35 \cdot 0.5055 \cdot 0.14]}{(4 \cdot 0.5055 - 3 \cdot 0.53) \cdot 0.5055 + 4 \cdot 0.14^2} = \frac{-3.20158 \times 10^{-3} + 3.13168 \times 10^{-3}}{0.218376 + 0.0752953} = \frac{-0.0699 \times 10^{-3}}{0.293671} = -0.00024$$

$$\dot{x}_0 = -\frac{wx_0(4 - 3\cos wt) + 2(1 - \cos wt)\dot{y}_0}{\sin wt} = \frac{0.35 \times 10^{-3} \times 110 \times 1.4116 + 2 \times 0.1372 \times 0.00024}{0.5055} = \frac{0.01147864}{0.5055} = 0.0227 = \dot{x}_0^+$$

$$\dot{z}_0 = -\dot{z}_0 \cdot w \cdot \frac{1}{\tan wt} = -0.0012 = \dot{z}_0^+$$

$$x(t) = \frac{\dot{x}_0}{w} \sin wt - \left(3x_0 + \frac{2\dot{y}_0}{w}\right) \cos wt + \left(4x_0 + \frac{2\dot{y}_0}{w}\right)$$

$$\dot{x}(t) = \dot{x}_0 \cos wt + \left(3x_0 w + 2\dot{y}_0\right) \sin wt = 0.0227 \times 0.8628 + (3 \times 110 \times 0.35 \times 10^{-3} + 2 \times -0.00024) \times 0.5055 = 0.0195556 - 0.0125035 = 0.0070521 = \dot{x}_f^-$$

$$y(t) = \left(6\dot{y}_0 + \frac{4\dot{x}_0}{w}\right) \sin wt + \frac{2\dot{x}_0}{w} \cos wt - (6wx_0 + 3\dot{y}_0)t + \left(y_0 - \frac{2\dot{x}_0}{w}\right)$$

$$\dot{y}(t) = (6\dot{y}_0 w + 4\dot{x}_0) \cos wt - 2\dot{x}_0 \sin wt - (6w)x_0 + 3\dot{y}_0$$

$$= (-660 \times 0.35 \times 10^{-3} + 4 \times -0.00024) \times 0.8628 - 2 \times 0.0227 \times 0.5055 + (6 \times 110 + 3 \times -0.00024)$$

$$= -0.04447 \times 0.8628 - 0.0224447 + 0.0423$$

$$= -0.042682 - 0.0224447 + 0.0423 = -0.0163 = \dot{y}_f^-$$

$$z(t) = \frac{\dot{z}_0}{w} \sin wt + z_0 \cos wt$$

$$\dot{z}(t) = \dot{z}_0 \cos wt - z_0 w \sin wt = -0.0012 \times 0.8628 - 10 \times 0.35 \times 10^{-3} \times 0.5055 = -0.0014 = \dot{z}_f^-$$

$$\begin{aligned} \Delta \vec{V}_i &= (\dot{x}_0^+ - \dot{x}_0, \dot{y}_0^+ - \dot{y}_0, \dot{z}_0^+ - \dot{z}_0) \quad i, j, k \\ &= (0.0227 + 0.01, -0.00024 - 0.02, -0.0012 - 0.001) \\ &= (0.0327, -0.02024, -0.0022) \end{aligned}$$

$$\begin{aligned} \Delta \vec{V}_f &= (\dot{x}_f - \dot{x}_f^-, \dot{y}_f - \dot{y}_f^-, \dot{z}_f - \dot{z}_f^-) \quad i, j, k \\ &= (0 - 0.0071, 0 + 0.0163, 0 + 0.004) \\ &= (-0.0071, 0.0163, 0.004) \end{aligned}$$



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$$\ddot{x} - 2\dot{y} - x = -\frac{(1-M)(x-M)}{r_1^3} - \frac{M(x+1-M)}{r_2^3} \rightarrow x = \frac{0.98877x(-0.48877-0.0123)}{\sqrt{(-0.48877-0.0123)^2+0.5^2}} + \frac{0.0123x(-0.48877+0.98877)}{\sqrt{(-0.48877+0.98877)^2+0.5^2}}$$

$$\ddot{y} + 2\dot{x} - y = -\frac{(1-M)y}{r_1^3} - \frac{My}{r_2^3} \rightarrow 0=0$$

$$\ddot{z} = -\frac{(1-M)z}{r_1^3} - \frac{Mz}{r_2^3} \rightarrow 0=0$$

$$r_1 = \sqrt{(x-M)^2 + y^2 + z^2}$$

$$r_2 = \sqrt{(x+1-M)^2 + y^2 + z^2}$$

$$M = 0.0123$$

$$1-M = 0.98877$$

midway 점  $\rightarrow M=0.5 \rightarrow x = -0.48877$

$$\dot{x} = \dot{y} = \dot{z} = 0$$

①  $[-0.5] \neq 3.90$

②  $[0=0]$

③  $[0=0]$

이 지점은 equilibrium 하지 않다.

현재 i 방향으로 3.9-0.5 의 차이만큼 3.4 로 가속도가 존재하기에 그 반대로 -3.4 i 의 힘을 가해 총합 0 을 만들어 평형이 되게 하자.

$$\ddot{x} = x - \frac{(1-M)(x-M)}{r_1^3} - \frac{M(x+1-M)}{r_2^3} = |3.9| - |0.5| = 3.4$$

$\therefore 3.4 - 3.4 = 0$

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④

$$\Delta \dot{x} = A \Delta x$$

$$\Delta x(0) = \Delta x_0$$

Let  $\lambda = \pm j\omega$  and  $(\beta, \bar{\beta})$  are 각각 eigenvalue and eigenvector.

Describe how to find IC that compose Halo orbit with single frequency  $\omega$ .

The solution of  $\Delta \dot{x} = A \Delta x$  is

$$\Delta x = e^{At} \Delta x_0 = a_1 \beta e^{j\omega t} + a_2 \bar{\beta} e^{-j\omega t}$$

$$= a_1 \beta (\cos \omega t + j \sin \omega t) + a_2 \bar{\beta} (\cos \omega t - j \sin \omega t)$$

$$= \cos \omega t (a_1 \beta + a_2 \bar{\beta}) + \sin \omega t j (a_1 \beta - a_2 \bar{\beta})$$

since  $\Delta x(0) = \Delta x_0$ ,  $\Delta x_0 = a_1 \beta + a_2 \bar{\beta}$

also  $\Delta \dot{x}_0 = j\omega (a_1 \beta - a_2 \bar{\beta})$

Solve for  $(a_1, a_2)$

If  $a_1 = a_2 \rightarrow \Delta x(0) = 2a_1 \operatorname{Re} \beta$

$a_1 = -a_2 \rightarrow \Delta x(0) = 2a_1 \operatorname{Im} \beta$

$\Rightarrow x(t)$  is periodic with  $\omega$ .

$\sum, \text{ If } x(0) \in \operatorname{span}(\operatorname{Re} \beta, \operatorname{Im} \beta).$

then the sol is periodic with  $\omega$ .